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Efficient Nitrogen Management Technologies for Sustainable Rice Production

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Abstract

The use of nitrogen (N) fertilizers in rice fields as a source of nutrition is the major source of emission of nitrous oxide (N₂O). Two key factors which control the flux of N₂O at the field level are the amount of N supplied and the efficiency at which it is absorbed by plants. To reduce the N₂O emissions, optimum N fertilizer application (in terms of input rate and time of application) and ideal fertilizer selection are crucial. Optimizing N-use efficiency (NUE) is crucial to sustain productivity and profitability. Depending on edaphic and climatic conditions, improved N management can dramatically cut greenhouse gases (GHGs) emissions. Producers must ensure that the kind, rate, and time of N application do not result in substantial losses owing to volatilization, leaching, or denitrification. Adoption of best nitrogen management practices like deep placement of urea, use of nitrification inhibitors, urease inhibitors, and slow-release nitrogen fertilizers will reduce the N loss and increase NUE. The goal of this review is to discuss in detail the various technologies that have been developed and refined to improve NUE and protect the environment.

Keywords: Nitrogen, Nitrification inhibitor, Urease inhibitor, Slow-release fertilizer, Brown manuring, LCC, GreenSeeker, SPAD

Introduction

Fertilizers boost agricultural productivity and encourage crop CO₂ uptake and decrease the need to cultivate new land (deforestation), resulting in fewer GHG emissions as a result of land use change. Nitrogen (N) is the most important element for the overall growth and development of rice plants (Subramanian *et al.*, 2020). The atmospheric N is not readily available to rice plants despite its high abundance in the air (around 79%). The proportion of fertilizer N in the total N input for crop production in India is increasing since the advent of the Green Revolution in the mid-1960s, but NUE has declined from 48 to 35% in 2018. There is a limited opportunity to achieve significant yield gains by applying more fertilizer N. Although optimal fertilizer use on agricultural crops reduces soil erosion, repeated applications of high N fertilizer doses may cause soil acidity, a negative soil health trait (Nayak *et al.*, 2020). Site-specific management

strategies based on the principles of synchronizing crop N demand with N supply from all sources, including soil and fertilizer, have the potential to ensure high yields while also preserving soil health (Vijayakumar *et al.*, 2021). Soil organic matter (SOM) is the repository for soil N. Balanced nutrient application and integrated nutrient management using organic manures and mineral fertilizers also contributed to the preservation and improvement of soil health (Nayak *et al.*, 2020). Thus, fertilizer N, when applied in a balanced proportion to other nutrients and in conjunction with organic manures, if available to the farmer, maintains or improves soil health rather than being detrimental (Nayak *et al.*, 2022). The good soil structure improves NUE and reduces N₂O losses. The challenge ahead is to manage N fertilizers in such a way that not only food demands are met continuously, but soil and environment remain healthy to support adequate food production with minimal environmental impact (Gobinath *et al.*, 2021).



N is added to the agricultural lands through inorganic N fertilizer which contains N in three chemical forms viz., ammonium (NH_4^+), nitrate (NO_3^-), and urea (Hakeem *et al.*, 2011). Globally, urea is the most preferred form of N in agriculture (Modolo *et al.*, 2015). However, once applied to the soil, urea undergoes three microbial-mediated transformations viz., hydrolysis, nitrification, and denitrification. Nitrification and denitrification are key processes contributing to N_2O emissions from the soil (Cameron *et al.*, 2013; Guo *et al.*, 2018). In the ammonia oxidation process, N_2O is produced by the chemical decomposition of hydroxylamine (NH_2OH). The loss of externally added N leads to economic and environmental implications. One potential way to mitigate N_2O emissions is to use nitrification and urease inhibitors to slow down the rate of nitrification and reduce the availability of the substrate (NH_4^+) for nitrification. The use of nitrogen fertilizer for crop production has an impact on soil health primarily through changes in organic matter content, microbial life, and acidity. Similarly, the production of N fertilizer also causes environmental pollution through the emission of GHGs. CO_2 emitted during ammonia synthesis and N_2O emitted during the production of nitric acid are the two most important GHG emissions connected with the manufacture of N fertilizers. Increasing NUE is critical for maintaining productivity and profitability. Improved N management, depending on edaphic and climatic conditions, can significantly reduce GHG emissions (Chatterjee *et al.*, 2019).

Farmers must ensure that the type, rate, and timing of N application do not cause significant losses due to volatilization, leaching, or denitrification (Cameron *et al.*, 2013; Vijayakumar *et al.*, 2021a). Good soil structure decreases N_2O losses. Best nitrogen management practices such as deep placement of urea, the use of nitrification inhibitors, urease inhibitors, and slow-release N fertilizers will reduce N loss and increase NUE (Vijayakumar *et al.*, 2021a). Blanket recommendations do not account for the spatiotemporal variability in soil N supply capacity (Subramanian *et al.*, 2020). Variable-rate fertilizer applicators in large fields are used in developed

countries to improve synchronization between crop N demand and N supply from various sources (Goud *et al.*, 2022). The goal of this paper is to go over in detail the various technologies that have been developed and refined to improve NUE and protect the environment. The management of fertilizer N has also been discussed in terms of providing adequate amounts of nutrients to crop plants and maintaining soil health.

Nitrification inhibitors

The microbial decomposition of N in soils, manures, and nitrogenous fertilizers produces N_2O , which is often exacerbated when available N exceeds plant requirements, especially in wet conditions. The use of NIs enhances NUE by extending the period of N available to the crop plants which leads to increased N uptake by crop plants due to the matching of soil available N with crop N demand (Huber *et al.*, 1977; Vijayakumar *et al.*, 2021a). Arresting nitrification could be a key strategy to improve N recovery and agronomic NUE in situations where the loss of N is significant. Nitrification inhibitors (NIs) selectively inhibit the microbial enzymes responsible for the conversion of NH_4^+ to NO_3^- . It reduces the risk of loss of N through leaching or denitrification and subsequently increases the NUE (Ruser and Schulz, 2015; Norton and Ouyang, 2019). The ammonia monooxygenase (AMO) is the first enzyme that is involved in the oxidation of NH_4^+ to NO in soils. The inhibition of the AMO by NIs directly decreases the nitrification rate and it reduces the NO concentration which serves as a substrate for denitrification. Hence, the two main pathways of N_2O production in soils are blocked or their source strength is at least decreased. NIs viz., nitrapyrin (2-chloro-6-trichloromethyl pyridine) or N-Serve, AM (2-amino-4-chloro-6 methyl pyrimidine), dicyandiamide (DCD), Ammonium thiosulphate (ATS), Thiosulphonyl triamide (ZPTA), terrazole (etridiazole) and CMP (1-carbamoyl-3-methylpyrazole) slow down the nitrification process in soil and lower N_2O emissions by 10–15 percent (Malla *et al.*, 2005). However, few studies showed even a 30 to 50% reduction in N_2O emission (Sanz-Cobena *et al.*, 2017). The recommended dose of NI is 0.2–0.6 kg ai/ha. A synthetic NI viz., Nitrapyrin reduces

nitrate leaching, N_2O emissions, improves NUE, crop yields, and N uptake (Woodward *et al.*, 2021). Dicyandiamide (DCD) is another effective NI found more suitable for the temperate region. DCD was found less effective if the temperature is above 20°C because of its rapid decomposition. NIs are effective in inhibiting the emissions of environmentally harmful N compounds from agriculture into the soil, water, and air. The use of NI in paddy soils leads to increased grain yields by 19%, N-recovery efficiency by 30–40%, and reduced N_2O emissions by 73% (Lan *et al.*, 2013; Gaihre *et al.*, 2020). The NIs viz., nitrapyrin and dicyandiamide (DCD) are the most effective inhibitors of nitrification/denitrification for the period of 2–6 and 12–14 weeks, respectively (Delgado and Follett, 2010). Dimethypyrazole phosphate (DMPP) is also effective in increasing soil NH_4^+ -N content when combined with urea, organic and inorganic fertilizers and lower soil N_2O emissions in temperate environments (Yang *et al.*, 2016). However, there is little evidence of its efficacy in sub-tropical or tropical environments where temperatures and rainfall intensities are typically higher (Rose *et al.*, 2017). The application of urease inhibitors and NIs significantly

reduced inorganic N leaching (48%), N_2O (44%), and NO emission (24%) (Burzaco *et al.*, 2014; Qao *et al.*, 2015; Thapa *et al.*, 2016) while increasing crop yield (7.5%) and NUE (12.9%) (Abalos *et al.*, 2014). The inhibitor decreased the potential denitrification rate (PDR) at the rice heading stage but had little effect on the denitrifier gene abundance except for nitrapyrin, which decreased the *nirK* gene abundance (Meng *et al.*, 2020). The list of NIs which are synthetically made and used in agricultural practices is presented in **Table 1**. Although several synthetic NIs are found very effective in inhibiting nitrification, their uses in agricultural land are limited due to high cost, limited availability, adverse influence on beneficial soil microorganisms, and above all, poor extension and promotional activities. Only a few inhibitors have got approval for commercial marketing. The increase in the cost of fertilization could be counterbalanced by an increment in crop productivity. Also, the potential improvement in crop NUE could minimize the rate of external N fertilizer application by reducing the losses, and thereby lowering fertilization costs (Abalos *et al.*, 2014).

Table 1. Common synthetic nitrification inhibitors

N-Source	Base Compound	N-Process	Common Names	N-Content	Inhibition Duration (weeks)
Nitrapyrin	2-chloro-6-trichloromet hylypyridine	Nitrification, denitrification	N-serve, stay-n 2000	12	2–6
DCD	Dicyandiamide	Nitrification	DCD, Ensan	1.6	4–8
DMPP	3,4-dimethypyrazoazole phosphate	Nitrification	Entec , Dmpp	12–26	6–8

Source: Havlin *et al.*, (2014)

Natural Nitrification Inhibitors (NNIs)

Natural NIs also known as botanical NIs encapsulate control water entry and rate of dissolution by providing a protective cover to the conventional soluble fertilizer which makes N release and availability more synchronized with plant requirements (Abbasi *et al.*, 2011). It also helps in improving soil health by reducing nitrification and N_2O emissions and enhancing crop productivity (Banik *et al.*, 2016). The usage of natural NIs like neem cake improves the N

recovery efficiency of applied N in arable soil (Hala *et al.*, 2014). The NNI like neem oil can inhibit the nitrification rate up to 20–50% in the soil, which is slightly lower than that of synthetic NIs like DCD (56–80%) (Raza *et al.*, 2019). Another potential natural NI is Karanj (*Pongamia pinnata*) seed extract which minimizes N_2O emission from soil (Banik *et al.*, 2016). It acts as a highly efficient NI (62–75% reduction in nitrification) as well as an N_2O mitigator (92–96% reduction in N_2O emission)



(Majumdar, 2002). The seed cake and extracts of Mahua contain alkaloids called saponin which slow down the N mineralization through nitrification inhibition. Based on the incubation study conducted on clay loam soil Kumar *et al.*, (2015) found that the nitrification inhibitory effect of mahua cake extract persisted only for 20 days. The advantages of NNI are easily available, cheap, and eco-friendly (Upadhyay *et al.*, 2011). Some natural NIs obtained from the different plants are enlisted in **Table 2**. The drawbacks of synthetic NIs like high cost, limited availability and adverse effect on beneficial soil microbes are solved through NNI. Therefore, it is necessary to develop and promote plant-based NIs (natural NIs) for augmenting NUE, crop productivity, and for safeguarding the environment.

While discovering a new NNI it is important to look for the following things. (i) *Specificity*: It should

block the conversion of ammonium to nitrate, i.e. the activity of *Nitrosomonas*, and be non-toxic to other soil organisms, animals, and humans. (ii) *Persistence*: The material should stay active in the soil for an adequate period. Compounds subject to rapid degradation will not be useful. (iii) *Mobility*: It should move with the fertilizer and nutrient solution. Compounds with too high vapour pressure may move too fast and compounds easily absorbed are probably not very effective (iv) *Economy*: The chemical should be cheap as it is used as an additive to fertilizers (Slangen and Kirchhoff, 1984). In India, 100% of urea produced is neem (*Azadirachta indica*) oil coated. The chemical compound present in neem oil act as a nitrification inhibitor (NI) and also act as a physical barrier thereby slowing down the speed of urea solubility (Reddy and Prasad, 1975).

Table 2. Natural nitrification inhibitors

Common Name	Scientific Name	Alkaloids	Reference
Neem	<i>Azardicta Indica</i>	Azaridictin	Slangen and Kerkhoff (1984)
Karanj	<i>Pongamia Glabra</i>	Karanjin, Glabrin, glabrosaponin	Modolo <i>et al.</i> , (2015)
Mahua	<i>Madhuca longifolia</i>	Saponin	Bisht <i>et al.</i> , (2018)

The ideal conditions where the use of NI is recommended

- Use of NIs may be advantageous in situations where the loss of N due to leaching and denitrification is accompanied by nitrification of fertilizer nitrogen.
- NIs are more effective in light-textured soils so their use may be more effective under these soil conditions as the effectiveness of these compounds decreases fast in heavy-textured soils.
- The application of NIs should be confined to the soil microsites where nitrification occurs rather than treating the entire soil volume so that the concentration of these compounds could be high enough for a reasonable period (Sahrawat and Mukerjee, 1977).

Urease Inhibitors (UIs)

Upon addition of urea to wet soil, it undergoes hydrolysis by the enzyme urease to generate ammonium carbonate, which is more prone to ammonia volatilization loss as carbonate increases the pH in the vicinity (Sahrawat, 1980). Urease enzyme is found both in the soil as well as in plant residues. UIs are chemical compounds that block the activity of the urease enzyme and reduce the rate of hydrolysis of urea to ammonium thereby it reduces the N loss through ammonia volatilization when urea is surface applied (Horta *et al.*, 2016). UIs gradually slow down the hydrolysis of urea for a period of 7 to 14 days by suppressing the activity of urease. The commonly known UIs are N-(n-butyl) thiophosphoric triamide (NBPT), and N-(n-propyl) thiophosphoric triamide (NPPT), PPD/PPDA (phenyl phosphorodiamide), TPT (tiophosphoryl triamide), PT (phosphoric triamide), HQ (hydrquinone). NBPT is sold in the

trade name of Agrotain and Limus is new UI that contains two active ingredients (NBPT and NPPT). Among the numerous forms of UI, NBPT has seen the maximum commercial application (Sanz-Cobena *et al.*, 2008; Abalos *et al.*, 2014). UIs can reduce N_2O emissions by up to 80 percent (Sanz-Cobena *et al.*, 2017). UIs, can only be used in conjunction with urea or urea-containing fertilisers (including organic sources). Many factors like soil pH, the texture of soil, and N application rate influence the efficiency of UIs. The hydrolysis of urea is rapid in high soil PH, or soil which is poorly buffered against an increase in pH. Thus, among the soil type, in alkaline soils, the efficiency of UIs is found to be highest. Similarly, in coarse-textured soils and at high N fertilization rates, the efficiency is higher (Abalos *et al.*, 2014). Most of the inhibitors including NBPT are highly effective in neutral soil with a moderate amount of organic matter.

Urea treated with NBPT reduces NH_3 loss by around 53% and yield is gained by 6.0% and varies from 0.8 to 10.2% depending on crop species (Cantarella *et al.*, 2018). UIs like NBPT have the potency to reduce ammonia volatilization and nitrite (NO_2) accumulation in the soil by altering the kinetic and thermodynamic behavior of the urease enzyme (Pan *et al.*, 2016). Thiousulfates can be used as UIs to reduce ammonia volatilization from urea or urea ammonium nitrate (UAN) fertilizer. Urea containing Ammonium Thiosulfate (ATS) has been shown to reduce NH_3 volatilization losses up to 11% as compared to UAN (Solan and Anderson, 1995). ATS by itself or in association with urea did not affect the soil microbial biomass pool. On the other hand, a field experiment performed with Canadian clay

loam and fine sandy loam soils showed inconsistent results concerning urease inhibition by ATS (Modolo *et al.*, 2018). Inhibitor N-(n-propyl) thiophosphoric triamide (NPPT) has shown a similar advantage of reducing ammonia volatilization losses from urea, and application of NBPT+NPPT mixture reduced NH_3 volatilization losses by 6% as compared to NH_3 losses of up to 25% in control (Li *et al.*, 2017; Hull, 2018). The application of 12 kg Hydroquinine (HQ) on alluvial soil in conjunction with 120 kg Urea-N ha^{-1} , decreased N_2O emission by 5% in rice and 7% in wheat systems as compared to the crops grown solely in the presence of 120 kg urea N ha^{-1} (Modolo *et al.*, 2018). Reduction in N_2O emission on the application of UIs along with urea ranged from 5% with hydroquinone to 31% with thiosulphate in rice. Contrary to the earlier finding, Malla *et al.*, (2005) reported the combined application of UI (Hydroquinone and thiosulphate) and urea increased N_2O emission as compared to the application of urea alone. However, the global warming potential (GWP) was lower with the inhibitors (except hydroquinone) as compared to urea alone (Malla *et al.*, 2005). In rice, the application of NBPT both in no-till and conventional tillage reduced the ammonia volatilization by delaying the conversion of N to NH_3 . However, the magnitude of the effectiveness of inhibitors was associated with soil, season, climate, and cultivation system (Marchesan *et al.*, 2013). Another study conducted at IRRI, Philippines revealed the use of NBPT improve seed germination (from 9.32 to 16.22% for Apo and from 17.76 to 36.81% for Hanyou3) and plant growth and reduced ammonia volatilization (Qi *et al.*, 2012). The properties of various synthetic UIs are presented in **Table 3**.

Table 3. Properties of synthetic urease inhibitors

Source	Common Names	Base Compound	N Process	N Content (%)	Inhibition duration (Weeks)
NBPT	Agrotain, Super U	N-(n-butyl) thiophosphoric triamide	Volatilization	46	2 to 3
Thiousulphate	ATS, CaTS	Ammonium or Calcium thiosulphate	Volatilization, Nitrification	12	2 to 3
NPPT	Limus	N-(n-propyl) thiophosphoric triamide	Volatilization		
Hydroquinine	HQ	Hydroquinine	Volatilization		-

Source: (Havlin *et al.*, 2014)

Natural Urease Inhibitors (NUIs)

These inhibitors are naturally found and obtained from plant parts and these chemical compounds block the activity of the enzyme urease thereby reducing the leaching losses. It has the potential to retard the loss of urea from agricultural soil and thus it may be used along with urea for improved utilization of the applied N by plants (Mathialagan *et al.*, 2017). The NUIs obtained from various plant parts are presented in **Table 4**. Allicin, a plant derived inhibitor obtained from garlic (*Allium sativum* L.) has shown the potential to inhibit urease activity in the soil (Mathialagan *et al.*, 2017). However, its inhibition is about 75% lower than NBPT at steady state (Matczuk and Siczek, 2021). Tannin, a polyphenolic extract obtained from the bark of *Acacia decurrens* (Green wattle; Fabaceae) or seed coat of *Terminalia chebula* (Inknot; Combretaceae) inhibited both pure urease (urease tablets-BDH) and soil ureases to the same extent that did mercuric chloride and catechol, known urease inhibitors (Modolo *et al.*, 2015). Indeed, with urea-polyphenol mixtures, NH_3 volatilization from the soil surface decreased upon soil fertilization. These results highlight the potential of tannin like polyphenols from green wattle and inknot as potent urease inhibitors (Fernando and Roberts 1976). In addition, some natural products such as phenolic compounds (methyl gallate, stilbenoids, and flavonoids) can suppress urease efficiency (Hussain *et al.*, 2021).

Table 4. Natural urease inhibitors

Inhibitors	Obtained from	Reference
Allicin	Garlic	Matczuk and Siczek (2021)
Tannin (polyphenolics)	<i>Acacia decurrens</i> (Green Wattle)	Modolo <i>et al.</i> , (2015)
Quercetin	<i>A. cepa</i>	Modolo <i>et al.</i> , (2015)

Slow/Controlled Release Nitrogen Fertilizers

This is a granulated fertilizer that differs from regular fertilizers by releasing nutrients slowly or gradually

into the soil. The fertilizer contains a plant nutrient in a form that extends its availability for plant uptake significantly longer than a reference fertilizer such as ammonium nitrate or urea, ammonium phosphate is commonly known as a slow-release fertilizer. Slow-release N fertilizers extend the period of N available to the crop plant by discharging the soluble N (NH_4 and NO_3) over several weeks/months and increase the amount of fertilizer uptake by the plant through synchronizing plant nutrient demand and soil N availability. This type of fertilizer is not readily water-soluble, which means it dissolves more slowly thereby it increases NUE and decreasing nutrient loss.

The demand for N increases gradually from germination to flowering. Usually, young plants have little demand while the demand for N increase from active tillering to the milking stage. The use of slow-release N-fertilizer ensures slow release of N to match crop demand. Slow-release N-fertilizers extend the period of N availability to crop plants as they release the N gradually and steadily in the soil solution thereby it increases NUE and decreasing its losses. The list of slow-release N-fertilizers is presented in **Table 5**. The slow-release N-fertilizers are classified into two categories viz., coated and uncoated. These products have been found to improve the recovery of applied N by 33% in cereal grains all over the world, and consequently decrease the external fertilizer applications rate. There are two types of slow-release N-fertilizers available in the market viz. coated (induced slow release) and uncoated products (inherently slow release).

Types of slow-release nitrogen fertilizers

Coated slow-release N fertilizers: The coated slow-release N fertilizers contain an external coating consisting of hydrophobic chemicals to provide a physical barrier against water. This type of fertilizer is not readily water-soluble, which means it dissolves more slowly. This promotes the gradual release of urea into the soil solution thereby it minimizes N losses and improves its uptake by crops (Akiyama *et al.*, 2010). The release of N is primarily controlled by the external barrier that surrounds the N. Thus, it releases the N rapidly once the barrier is removed.

Examples of coated products are neem-coated urea, sulphur-coated urea, and polymer-coated urea. In neem-coated urea, 0.5 kg of neem oil is used per tonne of urea. Polymer-coated fertilizers are the most recent technology for controlling N release and reducing N losses by leaching, denitrification, and volatilization. Polymer-coated multi-nutrient fertilizers supply all three fertilizer elements (NPK) which are essential for plant growth and development. These polymer-

coated fertilizers viz; Osmocote, Multicote, and Nutricote gradually release nutrients over extended periods (it can be shorter as three months and longer as eighteen months). Some commonly used coated N-fertilizers are listed in **Table 5**. The coated slow-release N fertilizers are comparatively cheaper than inherently slow-release N fertilizers as the products used for coating are easily available at low cost. In India, 100% urea manufactured is neem-coated urea.

Table 5. Coated N fertilizers

N-source	Base Compound	Common Name	N Content (%)	Inhibition Duration (Weeks)
Neem coated urea	Urea	NCU, NICU (Nimin-coated urea)	46	2-6
Polymer Sulphur-coated urea	Urea	Polyplus, Poly-S	38-42	6-16
Sulfur coated urea	Urea	Enspan, SCU	30-42	4-12
Polymer resin-coated urea	Urea	Polyon, Meister, Escote	38-44	8-14

Source: Havlin *et al.* (2014)

Inherently (uncoated) slow-release N fertilizers: Slow release is the inherent physical characteristic of uncoated products like isobutylidene diurea (IBDU) (31% N), urea form (35% N), and methylene urea (39-40% N) (Varadachari and Goertz, 2010). These are

slightly soluble in soil solution, where the N release rate depends on microbial activity and hydrolysis. The inherently slow-release N fertilizers along with their N content and inhibition period are presented in **Table 6**.

Table 6. Slow-release N-fertilizer compounds

N-source	Base Compound	Common Names	N Content (%)	Inhibition Duration (weeks)
Urea Formaldehyde	Urea forms, Methylol urea	Nitamin, Nitroform, Folocorn	35-40	6-10
Isobutylidene Diurea	Isobutylidine urea	IBDU	31	10-16
Triazone	Triazonefurea	N-sure	28-33	6-10
Melamine	2,4,6-triamino-1,3,5-triazine	Nitrazine	50-60	6-12
Crotolidene Diurea	Urea Crotonaldehyde	Crotodur, Triabon	34	6-12

Source Havlin *et al.* (2014)

Brown manuring

Generally, brown manuring is the practice of growing *Sesbania spp.* and rice together. When these *dhaincha* plants overtake the rice plants in height at about 25

days of co-culture, a broadleaf herbicide viz., 2, 4-D (selective herbicide) is applied to kill *Sesbania* plants, not the rice plants. After 4-5 days of herbicide spraying, *Sesbania* leaves will fall on the ground and



form mulch and help in smothering weeds. This is called the knocking down effect. The post-emergence herbicide spray on green manure leaves results in loss of chlorophyll in *Sesbania* leaves appear brown in colour and it is referred to as brown manuring.

Advantages of brown manuring

- Compete with weeds thus reducing their growth.
- Reduce the N requirement of plants as legumes fixed N from the atmosphere through bacteria present in their nodules.

- Prevent the loss of water due to evaporation and thus help in water conservation.
- Reduce the cost of cultivation by reducing the weed control cost and fertilizer N requirement.
- Increase soil organic carbon content and soil fertility.

The differences between green manure and brown manure are given in **Table 7**.

Table 7. Green manures VS Brown manures

Green Manures	Brown Manures
Moisture is necessary for incorporation and decomposition	Moisture is conserved during the practice
The risk of soil surface erosion is after incorporation	The plants are left standing to protect light texture soil from the risk of soil erosion
The microbial population is necessary for decomposition	Chemical desiccation will take place
It is the incorporation of a manure crop by tillage before seed set usually around flowering	It is a no-till version of green manuring, where herbicides are used to kill the manure crop and weeds

Source: Patil *et al.*, (2020)

Sesbania is a live cover that offers interference to weeds during the pre-killing period and later as a dead residue mulch (at the post-killing period) offers weed suppression and stimulates rice crop growth by the addition of organic matter and nitrogen release. The knocking down of *Sesbania* by 2,4-D application hastens the decomposition and release of nutrients present in *Sesbania* as compared to *in situ* incorporation. Also, brown manure crops are grown between the lines of rice crops and no free space is available for weeds to germinate and spread as a result a minimum weed population is recorded in

brown manuring. *Sesbania* could add C and N into the soil, which facilitates favourable microbial activity (Phukan and Bora, 2012). Other leguminous green manuring crops like sun hemp, cowpea, lentil, etc. are also potential brown manure crops for rice crops. Any pulse crop may be grown for brown manuring. Moreover, *Kharif* pulses which have good foliage and rapid growth are more suitable for this purpose. Nutrient content, Carbon-Nitrogen (C:N) ratio of green manure crops (**Table 8**), and the effect of brown manuring on soil organic carbon and post-harvest available N (**Table 9**) are highlighted below.

Table 8. Nutrient content and C: N ratio of major green manure crops

Crops suitable	Scientific name	Total N	C:N Ratio	Total P	Total K
Sun hemp	<i>Crotalaria juncea</i>	3.97	21:1	0.37	4.80
Dhaincha	<i>Sesbania aculeata</i>	1.90	44:1	0.34	3.60
Sesbania	<i>Sesbania speciosa</i>	2.71	40:1	0.53	2.21

Source: Iliger *et al.* (2017)

Table 9. Effect of brown manuring on soil organic carbon and post-harvest available nitrogen.

Year	Initial OC content of soil (%)	OC content after harvest (%)	% increase in organic carbon	Initial soil available nitrogen content (kg/ha)	Soil available N content after harvest (kg/ha)	% increase in soil available nitrogen
2014	0.54	0.69	0.15	283.0	320.2	13.7
2015	0.58	0.71	0.13	285.38	324.6	13.7
Mean	0.56	0.70	0.14	284.19	322.4	13.4

Source: Samant and Patra (2016). OC - Organic carbon.

More use of organic manures/green manures

The use of solid organic manure reduces the N_2O emission, however, it depends on the type of manure used (Webb *et al.*, 2010). While organic sources such as farmyard manure (FYM), green manure, and crop residues of rice and wheat increased the N_2O emission (Bhatia *et al.*, 2005). The incorporation of organic inputs, such as rice straw and green manure in rice soils promotes CH_4 emission (Van der Gon and Neue, 1995; Vijayakumar *et al.*, 2021). In Mediterranean systems use of solid manures significantly decreased N_2O emissions (23%) (Aguilera *et al.*, 2013) and has the potential to exacerbate long-term C sequestration (Ding *et al.*, 2012). Evidence from past experiments indicates that the technique of slurry application in agricultural soils is a crucial variable in regulating N_2O flux. Based on a meta-analysis study Hou *et al.*, (2015) reported that slurry injection could dramatically increase direct emissions as compared to broadcasting.

Manure, such as FYM, boosts CH_4 flux by providing organic carbon and nitrogen for microbial activities, as well as functioning as an electron source. In comparison to the application of a 100% recommended dose of N through urea, substituting 50% of inorganic N with FYM increased GHG emission by 172 percent (Pathak *et al.*, 2003). Crop residue incorporation/retention also influences the CH_4 flux by increasing the organic matter availability. The CH_4 flux increased from 100 to 500 kg ha⁻¹ yr⁻¹ with the increase of rice straw incorporation from 0 to 7 t ha⁻¹ (Sanchis *et al.*, 2012). The methane emissions were lowest in the unfertilized plot (28.4 kg ha⁻¹) and highest (41.3 kg ha⁻¹) when the total amount of N was applied by organic sources (Bhatia *et al.*, 2005). However, when

compared to FYM, biogas slurry lowered emissions by 2-3 times, indicating that biogas slurry should be favoured over FYM for reducing CH_4 emissions (Debnath *et al.*, 1996). Composting, incorporation of organic manures/ crop residues during the off-season i.e. drained period, and application of fermented manures like biogas slurry instead of unfermented farmyard manure reduce methane emission (Pathak and Wassmann, 2007) thus, promoting aerobic degradation of organic manure which reduces methane emissions.

Leaf colour chart (LCC)

The LCC is used to determine the N fertilizer needs of rice crops by determining the greenness of the rice leaf as it is highly influenced by N content. It is an inexpensive, small size and easy-to-use tool (Bhavana *et al.*, 2020). The use of LCC ensures the precise application of N fertilizer to rice crops. It has four green strips (four to six), with colour ranging from yellow-green to dark green. LCC is a substitute for the chlorophyll meter (SPAD) to estimate rice leaf N status. LCC readings are taken once a week until the first flowering, starting from 14 days after transplanting for transplanted rice and 21 days after seeding for wet direct seeded rice. The topmost fully expanded leaf from each hill is selected and leaf colour is compared by placing the middle part of the leaf on LCC. N top dressing is recommended whenever the green colour of more than 5 out of 10 leaves is found equal to or below the critical value. The critical value is 3 for varieties with light green foliage and 4 for all other varieties and hybrids. If the critical value of the leaf falls below the threshold value, 35 kg N/ha during *kharif/kar/kuruvai/navarai* and 30 kg N/ha during *rabi/samba/thaladi/pishanam* season need to be



applied. There is a considerable yield increase under LCC-based N management as N supply matches with the crop demand because of the timely supply of the optimum dose of N-fertilizer. The use of LCC leads to the saving of N fertilizers to the tune of 20 – 40 kg/ha (Sudhalakshmi *et al.*, 2008). Although it has many advantages over other N management tools, still it has a few limitations like sunlight influences the readings

if the measurements are not taken under the shade; deficiencies are identified after the symptoms are developed, and by this time crop might have been affected by the deficiency; LCC cannot give the exact values like the analysis done in the lab. Readings were taken in the morning (8-10 AM) under the shade of the body to avoid the influence of sunlight. The merits and demerits of LCC are presented below (**Table 10**).

Table 10. Merits and demerits of LCC tool

Merits of LCC	Demerits of LCC
LCC is an uncomplicated and easy-to-use tool for farmers to measure the nitrogen status of the leaf and to identify the instance for N top dressing.	LCC fails to specify minor variations in leaf greenness as the colour shades lie in between two shades.
LCC is cheap and portable thus, making it easy to carry to the field for estimating the N status of the leaf.	The comparative accuracy of LCC is relatively lower than the chlorophyll meter.
It is a non-destructive method and doesn't involve any laboratory analysis.	LCC developed for a particular region may not be appropriate for other regions. Similarly, the same LCC is not suitable for hybrid rice and HYV.
LCC can be better suited to a site-specific nutrient management approach.	LCC was used only to adjust the time of N top dressing not for basal N appliance.
Any specific knowledge or skill is not required for using LCC as it involves only comparing the leaf colour with a standard chart.	Though it does not require any specific skill to use, the user should be careful while taking the reading to avoid errors due to sunlight, time of observation, and selecting leaf for observation.

Source: Bhupenchandra *et al.*, (2021).

Integrated Nutrient Management (INM)

INM is the judicious use of all possible nutrient sources to meet the plant nutrient requirement at an optimum level to sustain the desired crop productivity with minimal impact on the environment. In INM, the immediate nutrient requirement of the crop is met through chemical fertilizers. Thus, the rate and time of chemical fertilizer application should synchronize with the real-time need of the crop. The slow and long-term release of nutrients from organic sources helps in meeting the long-term need of the crop. The goal of INM includes (i) Optimization of the benefits from all possible sources of plant nutrients in an integrated manner to achieve a given level of

crop production (ii) Maintenance of plant nutrient supplying capacity of soil to ensure sustainable crop productivity (iii) Ensuring higher nutrient use efficiency, minimization of nutrient loss and mitigation of harmful environmental impacts (iv) Minimizing the use of chemical fertilizers thereby reducing the cost of cultivation and enhancing profitability (Vijayakumar *et al.*, 2021a). The INM for different rice production systems is given below (**Table 11**).

Components of INM

Organic manures: Farmyard manure, compost, vermicomposting, biogas slurry, poultry manure, crop residues, and bio wastes like press mud, sugarcane baggages etc.

Green manures & Green leaf manures: Dhaincha (*Sesbania aculeata*), *Sesbania rostrata*, Sunhemp (*Crotalaria juncea*), *Pongamia globra*, *Leucaena leucocephala*, *Azadiracta indica* and all legume pulses except French bean.

Chemical fertilizers: Urea, Ammonium Sulphate, Ammonium Nitrate, Calcium Ammonium Nitrate (CAN), etc.

Table 11. The recommended INM for rice

S.No	Recommendation	Yield	Reference
1	100% recommended dose of N through green manure with 50 percent NPK	62.7 q ha ⁻¹	Bhandarin <i>et al.</i> , (1992)
2	50% of N through green manure and the remaining 50% through chemical fertilizers	7.3 t ha ⁻¹	Sharma and Subehia (2014)
3	Application of soil-based BGA biofertilizer at the rate of 10 kg ha ⁻¹ along with 90 kg urea	10 percent high yield	Mohanty <i>et al.</i> , (2019)

Green seeker

The Green Seeker is a hand-held optical reflectance sensor that uses active radiation from red and near-infrared bands independent of solar conditions. The sensor samples at a very high rate (approximately 1000 measurements per second) and averages measurements between outputs. This device delivers output viz., NDVI, and ratio vegetation index (RVI) directly using the sensor readings at a rate of 10 readings per second with a travel speed of 0.5 m s⁻¹. The integrated optical sensing and application system measures nitrogen status in the leaf and provides information on the right time, right place, and right amount of N application in real-time (Song, 2021). The sensor unit has self-contained illumination in both the red (656 nm with about 25 nm full-width half magnitude) and near-infrared (774 with about 25 nm FWHM) bands. Sensor readings (NDVI and RVI) were collected 0.5m above the rice canopy across each plot, except plot borders and the average values were used to represent each plot (Zhang *et al.*, 2017). Sensor readings were collected at five different stages viz., tillering, panicle initiation, booting, before heading, and heading stage. The original technology was developed for large farms; however, a small handheld version that costs (approximately Rs. 40,000) a fraction of the original technology is now commercially available (Yao *et al.*, 2012).

Soil Plant Analysis Development (SPAD) Meter

Soil Plant Analysis Development (SPAD) Meter or chlorophyll meter developed by Minolta Company is a simple, portable diagnostic tool that measures the greenness or relative chlorophyll content and is mainly used to identify the crop N status and relative chlorophyll contents (Yuan *et al.*, 2016). The SPAD meter measures the difference between the transmittance of red (650 nm) and infrared (940 nm) light through the leaf, generating a three-digit SPAD value (Uddling *et al.*, 2007). It enables users to measure potential photosynthetic activity quickly and easily, which is closely linked to leaf chlorophyll content, crop nitrogen status, and leaf greenness. SPAD readings indicate the plant N status and the amount of N to be applied. It is a non-destructive method of N status estimation thereby it saves time and money. SPAD readings are greatly influenced by the specific part of the foliage where the measurements are made, as chlorophyll is not evenly distributed along the leaf blade. Several factors such as plant growth stages, cultivars, specific leaf weight, leaf thickness, leaf position on the plant, measurement location on a leaf, environmental stress, and solar radiation could significantly affect chlorophyll meter readings (Yuan *et al.*, 2016). The chlorophyll meter is too costly (around one lakh rupees) which is very high for a small-scale farmer. Leaf area-based N concentration



has a unique linear relationship with SPAD values of rice plants at all growth stages (Peng *et al.*, 1995).

SPAD values for different rice production systems are given below (Table 12).

Table 12. Critical SPAD values for different seasons, cropping conditions, and rice varieties

Crop establishment	Varietal group	Panicle density (m ²)	SPAD value	
			Dry season	Wet season
Transplanted rice	Traditional improved local aromatic rice	300-400	30-32	30
	Semi-dwarf indica varieties	400-500	32-35	35-37
	Hybrid rice	400-500	32-35	35-37
Broadcast sown	All varieties	High -800	29-30	30
		Medium-400 to 500	32	35
Drum seeded	All varieties	High 600-650	32	32
		Medium 400-500	32-35	32-35

Source: Balasubramanian *et al.* (2000)

4R nutrient stewardship-based N application

Any technology which ensures a more precise application of N fertilizer based on soil, plant, and field characteristics will increase the NUE and reduce the N loss. 4R nutrient stewardship-based N application involves applying the right dose, right time, right source, and right place enhances NUE (Vijayakumar *et al.*, 2021). For example, the demand-driven application of N by using a leaf colour chart (LCC) reduced N₂O emission and GWP by about 11% (Bhatia *et al.*, 2010) thereby synchronizing the timing of N application with plant N demand and reducing N losses, including N₂O emissions. It also helps in saving fertilizer costs due to the saving of input N rate (Surjandari and Batte, 2003). Accurate estimation of external N requirements by considering indigenous supply, and target yield will reduce N loss by avoiding the excess N application and subsequent direct and indirect N₂O emissions, while saving energy and lessening other GHG emissions (e.g., associated with manufacturing N fertilizers). The optimized N application might cut N₂O flux by up to 50 percent compared to non-optimized practices in both irrigated and rain-fed Mediterranean agroecosystems (Sanz-Cobena *et al.*, 2017). However, multiple studies have found that direct N₂O emission is non-linear in response to N intake (Philibert *et al.*,

2012; Kim *et al.*, 2013; Shcherbak *et al.*, 2014), and other factors, such as cultural operations, method of fertilizer application, time of application, source of N fertilizer and climate plays a major role in direct N₂O emissions (Aguilera *et al.*, 2013). For rice and wheat, three split applications of N were found more efficient than two split applications. Several findings revealed that choosing the correct fertilizer could help reduce emissions. The use of nitrate (NO₃) based fertilizers significantly lowered the N₂O emissions than ammonium-based fertilizers (Bouwman *et al.*, 2002). N is mostly broadcast applied in India and other Asian countries. The broadcasting of urea and the ammonium-containing fertilizers is often associated with higher volatilization losses and it can be largely reduced by incorporating urea into the soil. This is done in the case of dry direct seeded rice and wheat in IGP regions. The use of seed cum fertilizer drills also enables incorporation of urea into the soil and this method is gaining importance in IGP for sowing zero-till wheat in RWCS.

Time of Application of Nitrogen Fertilizer

For better rice crop growth and development, the perfect time is required for the application of fertilizer. N is required in large amounts for rice plants and the Recommended Dose of Nutrients (RDN) application is advised to broadcast three times (1/3rd is applied

before planting, incorporated in dry soil; 1/3rd at the mid-tillering stage and 1/3rd at panicle initiation stage) throughout its growing season. The effect of different times and methods of N application on rice crop is presented in **Table 13**.

Table 13. Time of application of fertilizer for rice crop

S.No	Method	Times of application	Yield	Reference
1	Direct seeded upland rice	3 splits 1/2 at 20 days, 1/4 th at tillering, 1/4 th at panicle initiation	More than expected	Kaur and Kaur (2017)
2	Direct-sown rice under lowland conditions.	4 splits, 17% at 21 days after sowing, 33 at 35 DAS, 33% at panicle initiation and 17% at first flowering	4.18 t ha ⁻¹	Thilagavathi and Ramanathan (2005)
3	Direct-sown rice under lowland conditions.	4 splits, 1/6 th at 15 DAS, 1/3 rd at tillering, 1/3 rd at panicle initiation, 1/6 th at flowering	4.92 t ha ⁻¹	Sathiya and Ramesh (2009)
4	Aerobic rice	4 splits – 1/6 th at 15 DAYS, 1/3 rd at tillering, 1/3 rd at PI, 1/6 th at flowering recorded higher tillers	2.82 t ha ⁻¹	Sathiya and Ramesh (2009)

Deep placement of Urea

Under direct seeded rice (DSR), deep placement of urea reduced N₂O flux by 93% compared to broadcast urea and thereby increased NUE and grain yields (Gaihre *et al.*, 2020). This is most plausible as UDP might have stored much of the nitrogen as NH₄⁺ in an anaerobic zone for a long time, where nitrification is less likely due to the lack of O₂. As a result, both nitrification and denitrification emissions of N₂O and NO could be lowered. Furthermore, UDP minimizes N loss through other processes such as NH₃ volatilization and surface runoff (Rochette *et al.*, 2013).

Decision support tool

Many decision support tools are now available for managing nutrient supply in different cropping systems. For example, the Nutrient expert decision support system (DSS) developed by the International Plant Nutrition Institute (IPNI) gives site-specific recommendations (SSNR) for hybrid maize genotypes. DSS was found effective tool under both conservation and conventional production system (Kumar *et al.*, 2014, 2015a, 2015b). DSS provides SSNR even in the absence of soil test values. It needs information that is easily given by the farmer/user. Similarly, for rice RiceXpert developed by ICAR-National Rice Research Institute (NRRI), Cuttack

gives N recommendations to standing rice crops by capturing the N status of the plant. The farmer needs to take ten photos of standing rice crops randomly across the field using a smartphone. After uploading captured images, the output i.e. N recommendation immediately delivered to the farmer in terms of urea. Similarly, unmanned aerial vehicle (UAV) mounted sensors capable of detecting N stress in rice plants even before it produces visible visual symptoms (Vijayakumar *et al.*, 2020a). The spectral signature of multi-spectral and hyper-spectral sensors are highly correlated with the N status of the plant. However, at present, the higher cost of the UAV system hinders its application at the field level (Vijayakumar *et al.*, 2021b).

Conclusion

We have reviewed the field-specific N management strategies based on the leaf colour chart, chlorophyll meter (SPAD), and GreenSeeker for the need-based application of N fertilizers in rice. Chlorophyll meters (SPAD), GreenSeeker, and LCC have been standardized for applying N fertilizer to rice crops based on their needs. Farmers are increasingly using a simple and inexpensive LCC to practice field-specific N application in rice, which can increase agronomic efficiency by 5 to 16 kg grain per kg N over the



farmers' fertilizer practice. Nutrient decision support tools that are computer, mobile, or web-based can also help manage fertilizer N in rice on a field-by-field basis (Vijayakumar *et al.*, 2022). To achieve higher NUE, farmers in India must significantly improve fertilizer N management by adopting technological innovations and avoiding N applications greater than the crop's need. Adoption of site-specific N management strategies has great potential; however, adoption of technologically advanced N options such as controlled release N fertilizers and nitrification and urease inhibitors will be dependent on the benefit:cost ratio of their use in India.

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